# 26-BS-CH101 — Chemistry

| Course Code | 26-BS-CH101 | Category | Basic Science Course |
|---|---|---|---|
| Course Title | Chemistry | L – T – P | 3 – 0 – 0 |
| Credits | 3 | Contact Hours | 36 |
| Semester | First or Second (by group) | Assessment | Theory: CA 30 + ESE 70 |

## Prerequisites

Builds on the Higher Secondary Chemistry of WBCHSE, CBSE or ISC. Under the group rotation it is taken in one semester by one group and in the other by the second; the syllabus is identical. Assumed:

- Atomic structure, the quantum numbers and electronic configuration; the periodic table (CBSE XI Units II–III; WBCHSE).
- Chemical bonding — ionic, covalent and coordinate bonds, VSEPR and hybridisation (CBSE XI Unit IV).
- Thermodynamics, equilibrium, electrochemistry and kinetics (CBSE XI Unit VI; XII Units III–IV).
- A three-hour bridge on atomic structure, bonding and the mole concept is provided for students without Class XII Chemistry.

## Course Objectives

- To establish the quantum-mechanical picture of the atom and the molecule as the basis of the properties of materials.
- To develop the thermodynamic, kinetic and electrochemical reasoning used to predict whether and how fast a process occurs.
- To connect the structure of materials and of water to their engineering behaviour and treatment.
- To introduce the instrumental methods by which composition and structure are determined.

## Course Outcomes

On completion of the course the student will be able to:

| CO | Statement | Bloom |
|---|---|---|
| CO1 | Apply the quantum-mechanical and molecular-orbital picture to the structure and properties of molecules and materials. | L3 – Apply |
| CO2 | Apply thermodynamics, chemical and phase equilibrium and reaction kinetics to engineering processes and compute the quantities. | L3 – Apply |
| CO3 | Apply the principles of electrochemistry to cells, batteries and fuel cells, and explain and control corrosion. | L3 – Apply |
| CO4 | Relate the structure of engineering materials and the chemistry of water to their engineering use and treatment. | L4 – Analyse |
| CO5 | Select an appropriate spectroscopic or instrumental method for an analytical problem and interpret its output. | L3 – Apply |

## CO–PO Articulation Matrix

*Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).*

| CO \ PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | 1 | 1 | 1 | – | – | – | – | – | 1 |
| CO2 | 3 | 3 | 2 | 2 | 1 | – | – | – | – | – | 1 |
| CO3 | 3 | 2 | 2 | 2 | 2 | 1 | 2 | – | – | – | 2 |
| CO4 | 3 | 2 | 2 | 2 | 2 | 2 | 2 | – | – | – | 2 |
| CO5 | 3 | 2 | 1 | 3 | 2 | – | – | – | – | – | 2 |

## Course Plan — Module-wise Syllabus

| Module | Title | Contents | Hours |
|---|---|---|---|
| I | Atomic and Molecular Structure | The Schrödinger equation and the particle in a box as a model; atomic and molecular orbitals; molecular-orbital theory of diatomic molecules, bond order and magnetic behaviour; π-molecular orbitals of conjugated systems; band theory of solids in outline; crystal-field splitting and the colour of transition-metal complexes. | 8 |
| II | Thermodynamics, Equilibrium and Kinetics | The laws of thermodynamics in review; entropy, free energy and the criteria of spontaneity; chemical potential; chemical and phase equilibrium, the phase rule and simple phase diagrams; chemical kinetics — order, molecularity, the rate law, the Arrhenius equation and catalysis. | 8 |
| III | Electrochemistry and Corrosion | Electrode potential and the electrochemical series; the Nernst equation; galvanic and concentration cells; conductance; batteries — primary, secondary and the lithium-ion cell — and fuel cells; corrosion — mechanism, factors, and protection by coatings, cathodic protection and inhibitors. | 7 |
| IV | Engineering Materials and Water | Polymers — addition and condensation polymerisation, thermoplastics and thermosets, elastomers, conducting and biodegradable polymers; composites and nanomaterials in outline. Water — hardness and its determination, boiler troubles, softening by lime–soda, zeolite and ion-exchange methods, and municipal treatment. | 8 |
| V | Instrumental Methods of Analysis | The basis of spectroscopy and the Beer–Lambert law; ultraviolet–visible, infrared and, in outline, nuclear magnetic resonance spectroscopy; elementary chromatography; the accuracy and precision of a measurement and the reporting of a result with its uncertainty. | 5 |

**Cross-cutting notes.** Module III (corrosion) and Module IV (materials and water) are the points at which the course serves the materials, durability and water concerns of the Civil, Mechanical, Electrical and Leather disciplines and connects to Environmental Science. The concurrent Chemistry Laboratory (BS-CH191/291) reduces at least one result with its uncertainty.

## Text and Reference Books

- Manisha Aggrawal, Chemistry - I, Khanna Publishing House (AICTE Prescribed Textbook)
- P. C. Jain and M. Jain, Engineering Chemistry, Dhanpat Rai.
- P. W. Atkins, Physical Chemistry, Oxford University Press.
- Manisha Aggrawal, Engineering Chemistry, Khanna Publishing House
- B. R. Puri, L. R. Sharma and M. S. Pathania, Principles of Physical Chemistry, Vishal.
- S. S. Dara and S. S. Umare, A Textbook of Engineering Chemistry, S. Chand.
- Shashi Chawla, A Textbook of Engineering Chemistry, Dhanpat Rai.

## Suggested SWAYAM / NPTEL and Online Resources

- NPTEL — Chemistry: Principles and Applications / Engineering Chemistry, 12 weeks.
- NPTEL — Electrochemical Energy Systems / Corrosion Engineering, 12 weeks.
- Virtual Labs (Ministry of Education) — Physical, Inorganic and Analytical Chemistry laboratories.

**Question-paper note.** At least one Group C question is a numerical problem — a free-energy or equilibrium computation, a Nernst potential, a rate or activation-energy calculation, or a hardness determination. Structure-and-reasoning questions require explanation from the underlying bonding or thermodynamics.